Tools — MasterFetch

MasterFetch provides rich web content extraction, multi-engine search, and same-domain crawling with a content cache. Outputs carry envelope signals (`content_ok`, `page_type`, `next_action`, `source_type`, `is_official`, `content_age_days`, `is_stale`).

Tool	Description
`mf_fetch`	Rich URL/PDF extraction with envelope signals.
`mf_search`	Keyless multi-engine web search.
`mf_crawl`	Best-first same-domain crawl.
`mf_cache_clear`	Clear the content cache.
`mf_screenshot`	Capture a page as a screenshot. **Registered only when a drivable headless browser engine is present** (spec `openhands` FR-029); absent from the default build.
`mf_version`	Return integration version info.

Use cases: extracting article content, searching multiple engines in parallel, crawling a domain.

Visibility switch: `masterfetch` (see `docs/howtos/tool-visibility.md`).

`mf_search` runs DuckDuckGo, Brave, OpenAlex, and Wikipedia in parallel (plus optional LangSearch / Tavily / Perplexity / Exa / Serper engines when their keys are configured). `exclude_engines` removes named backends before any request is dispatched; naming every configured engine returns an explicit "all engines excluded" result. An optional `engine` restricts the search to a single backend.

Set `respect_robots: true` on `mf_fetch` / `mf_crawl` when unsure whether a site allows crawling; do not hammer the same domain with repeated calls.

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mf_fetch

Fetch any URL or PDF with automatic content extraction.

Arguments

Argument	Type	Required	Description	Typical value
`url`	string	one of `url`/`urls`	Single URL to fetch	`"https://example.com/article"`
`urls`	array	one of `url`/`urls`	Array of URLs for parallel bulk fetch (up to 8 concurrent)	`["https://a.com","https://b.com"]`
`format`	enum	no	`markdown` (default), `html`, `text`, `raw`	`"markdown"`
`css_selector`	string	no	Narrow extraction scope	`"article"`
`focus`	string	no	Query string for BM25 post-extraction filtering (no re-fetch)	`"memory safety"`

Argument	Type	Required	Description	Typical value
`max_content_chars`	integer	no	Maximum content characters (min 500)	`40000`
`offset`	integer	no	Character offset to resume pagination	`40000`
`include_links`	boolean	no	Classify outgoing links into citations/navigation/external	`false`
`respect_robots`	boolean	no	Check robots.txt before fetching	`false`
`cache_ttl`	integer	no	Cache TTL in seconds (`0` bypasses cache)	`3600`

Example:

```
mf_fetch url="https://example.com" format="markdown"
mf_fetch urls=["https://a.example.com","https://b.example.com"] max_content_chars=20000
```

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mf_search

Keyless multi-engine web search; engines run in parallel and results are merged, deduplicated, and ranked with `relevance_score`, `fetch_relevance`, and `engines_consensus`.

Arguments

Argument	Type	Required	Description	Typical value
`query`	string	yes	Search query	`"zero-copy parsing in rust"`
`engine`	enum	no	Restrict to one backend: `openalex`, `wikipedia`, `langsearch`, `tavily`, `perplexity`, `exa`, `serper`	—
`exclude_engines`	array	no	Engine names to drop before dispatch	`["openalex"]`
`site`	string	no	Restrict results to a domain	`"rust-lang.org"`
`exclude_sites`	array	no	Domains to exclude	`["pinterest.com"]`
`freshness`	enum	no	`day`, `week`, `month`, `year`	`"week"`
`max_results`	integer	no	Cap after merge/dedup (1–500)	`6`

Argument	Type	Required	Description	Typical value
`per_engine_results`	integer	no	Cap per engine before merge (1–200)	`75`
`page`	integer	no	Result page (0–10)	`0`

Example:

```
mf_search query="rust async patterns" max_results=10
mf_search query="rust async patterns" exclude_engines=["openalex"]
mf_search query="site:doc.rust-lang.org ownership rules" site="doc.rust-lang.org"
```

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mf_crawl

Best-first, same-domain crawl starting from a URL. Each page is returned as markdown with `content_ok` and `page_type` envelope signals.

Arguments

Argument	Type	Required	Description	Typical value
`url`	string	yes	Start URL	`"https://docs.rs/serde"`
`max_pages`	integer	no	Maximum pages to fetch	`10`
`max_depth`	integer	no	Maximum crawl depth from start URL	`2`
`max_total_chars`	integer	no	Total character budget across pages	`200000`
`deadline_ms`	integer	no	Time budget in milliseconds	`120000`
`focus`	string	no	Query string for scoring/filtering pages	`"tutorial"`
`sitemap`	boolean or `"auto"`	no	Use sitemap (`true`), use if available (`"auto"`), or pure BFS (`false`)	`false`
`discover_only`	boolean	no	Return discovered URL map without fetching content	`false`
`crawl_urls`	array	no	Second-phase selective crawl: fetch only this subset of URLs	—
`respect_robots`	boolean	no	Check robots.txt before fetching	`false`

Example:

```
mf_crawl url="https://doc.rustup.rs" max_pages=15 max_depth=2 focus="getting started"
```

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mf_cache_clear

Clear the masterfetch content cache.

Arguments

Argument	Type	Required	Description	Typical value
`all`	boolean	no	`true` wipes all entries; `false`/omitted purges only expired entries	`false`

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mf_screenshot

Capture a page as a screenshot image. This tool is registered **only when a drivable headless browser engine is present** (spec `openhands` FR-029): the default build compiles no driver, so the tool is absent from the advertised tool set and the capability is never offered when it cannot work. When a bundled driver is enabled (the opt-in `headless-browser` Cargo feature) and an engine binary resolves on `PATH`, the tool registers and returns a screenshot (or an honest error recommending `mf_fetch` for text-based extraction).

Arguments

Argument	Type	Required	Description	Typical value
`url`	string	yes	URL to capture	`"https://example.com"`
`width` / `height`	integer	no	Viewport size in pixels	`1280` / `800`
`full_page`	boolean	no	Capture the full scrollable page	`false`

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mf_version

Return the masterfetch integration version, the ragent version, and a brief description of the tool set. Makes no network calls and always succeeds.

Arguments: none.